Global Tuberculosis Burden Analysis

Interactive Tableau Dashboard Report

Data Visualization & Business Intelligence Project

Prepared Using Tableau Desktop

1 Introduction

Tuberculosis (TB) remains one of the most significant infectious diseases worldwide, affecting millions of individuals every year. The purpose of this project is to analyze global TB patterns using interactive Tableau dashboards and visual analytics techniques.

The analysis focuses on identifying:

- Global TB incidence trends over time
- Regional differences in TB burden
- Geographic distribution of TB cases
- Countries with the highest estimated incidence
- Relationships between TB detection and mortality

The dashboards were designed following storytelling principles, interactive exploration techniques, and visual design standards to provide meaningful insights into global TB trends.

2 Dashboard 1: Global TB Burden Overview

This dashboard provides a high-level overview of global TB patterns and summarizes key performance indicators related to TB incidence and detection.

2.1 Key Visualizations

- Global TB Trend Line Chart
- Regional TB Burden Bar Chart
- KPI Summary Cards

2.2 KPIs Used

- Total Estimated Cases
- Average Detection Rate
- Highest Burden Year

2.3 Key Findings

- Global TB incidence increased steadily from 1990 until the early 2000s.
- The year 2002 represented the highest recorded TB burden.
- African regions (AFR) experienced significantly higher TB incidence compared to other regions.
- European and American regions generally exhibited lower TB burden levels.

2.4 Dashboard Screenshot

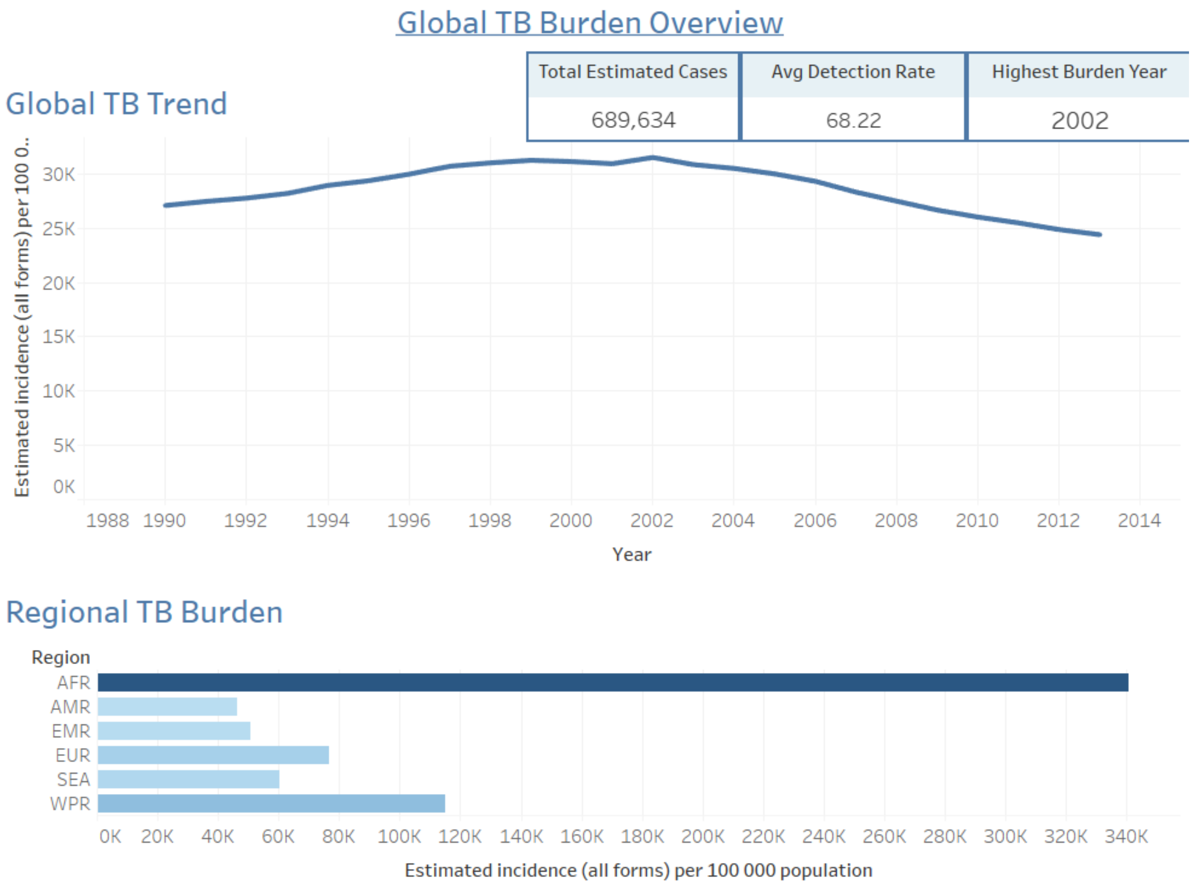

Figure 1: Global TB Burden Overview Dashboard

3 Dashboard 2: Geographic Distribution of Tuberculosis Cases

This dashboard explores the geographic distribution of TB burden across countries and regions using choropleth maps and country comparison charts.

3.1 Key Visualizations

- Geographic TB Distribution Map
- Top TB Countries Bar Chart

3.2 Filters and Interactive Features

- Region Filter
- Year Slider Filter
- Interactive Geographic Exploration

3.3 Key Findings

- African countries showed the highest TB incidence rates globally.
- Countries such as Namibia, Lesotho, Swaziland, and South Africa demonstrated particularly high TB burdens.
- European and North American countries generally displayed lower incidence levels.
- Geographic analysis highlighted unequal distribution of TB burden worldwide.

3.4 Dashboard Screenshot

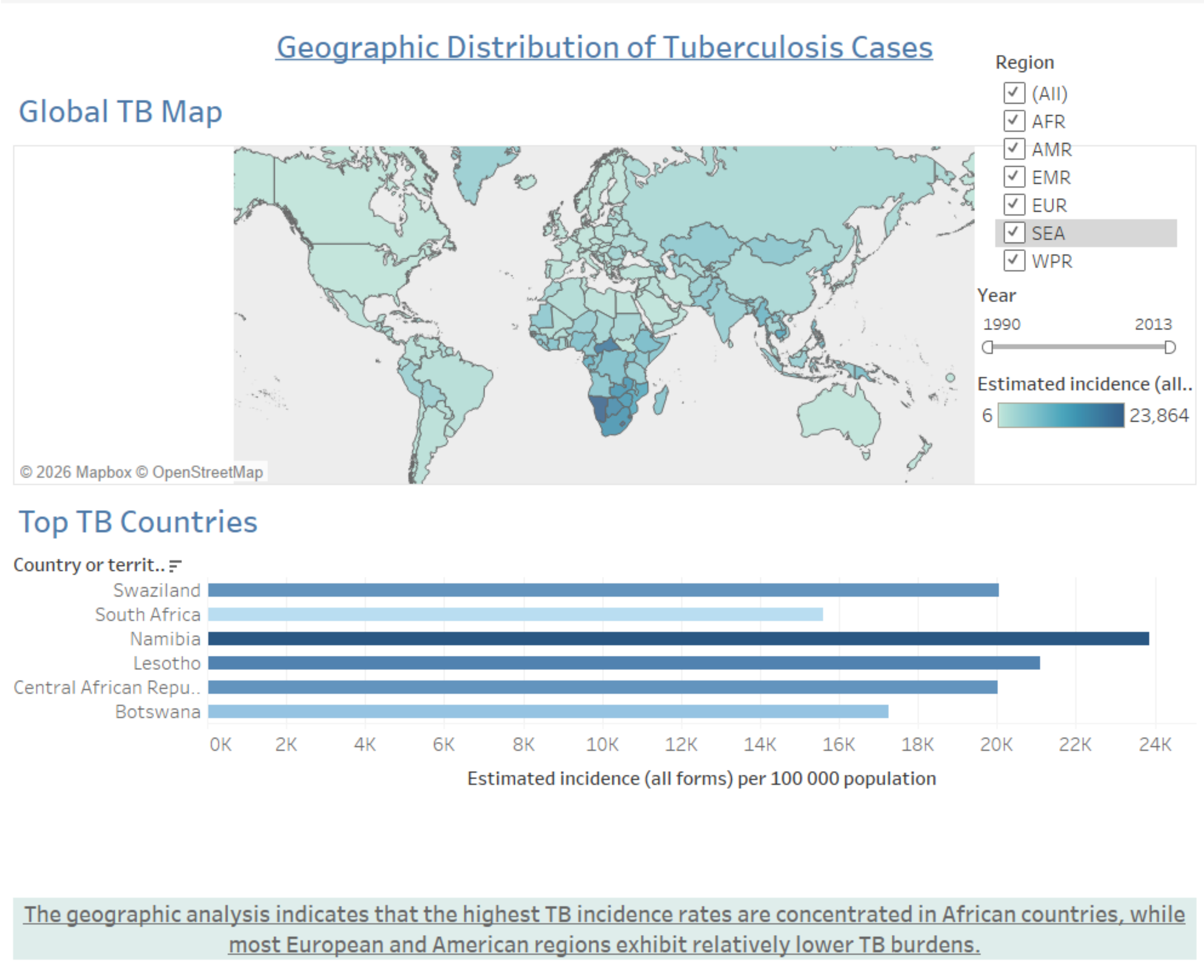

Figure 2: Geographic Distribution of Tuberculosis Cases Dashboard

4 Dashboard 3: Relationship Between TB Detection and Mortality

This dashboard investigates the relationship between TB detection rates and mortality using scatter plot correlation analysis.

4.1 Key Visualizations

- Scatter Plot Correlation Analysis
- Bubble Size Encoding by Estimated Incidence
- Region-Based Color Encoding

4.2 Filters and Interactive Features

- Year Filter
- Interactive Tooltips
- Region Legends

4.3 Key Findings

- Countries with higher TB mortality often exhibited lower detection performance.
- African countries generally showed larger TB burdens compared to other regions.
- Significant variability in TB outcomes exists across different regions.
- Larger bubbles represented countries with higher estimated TB incidence.

4.4 Dashboard Screenshot

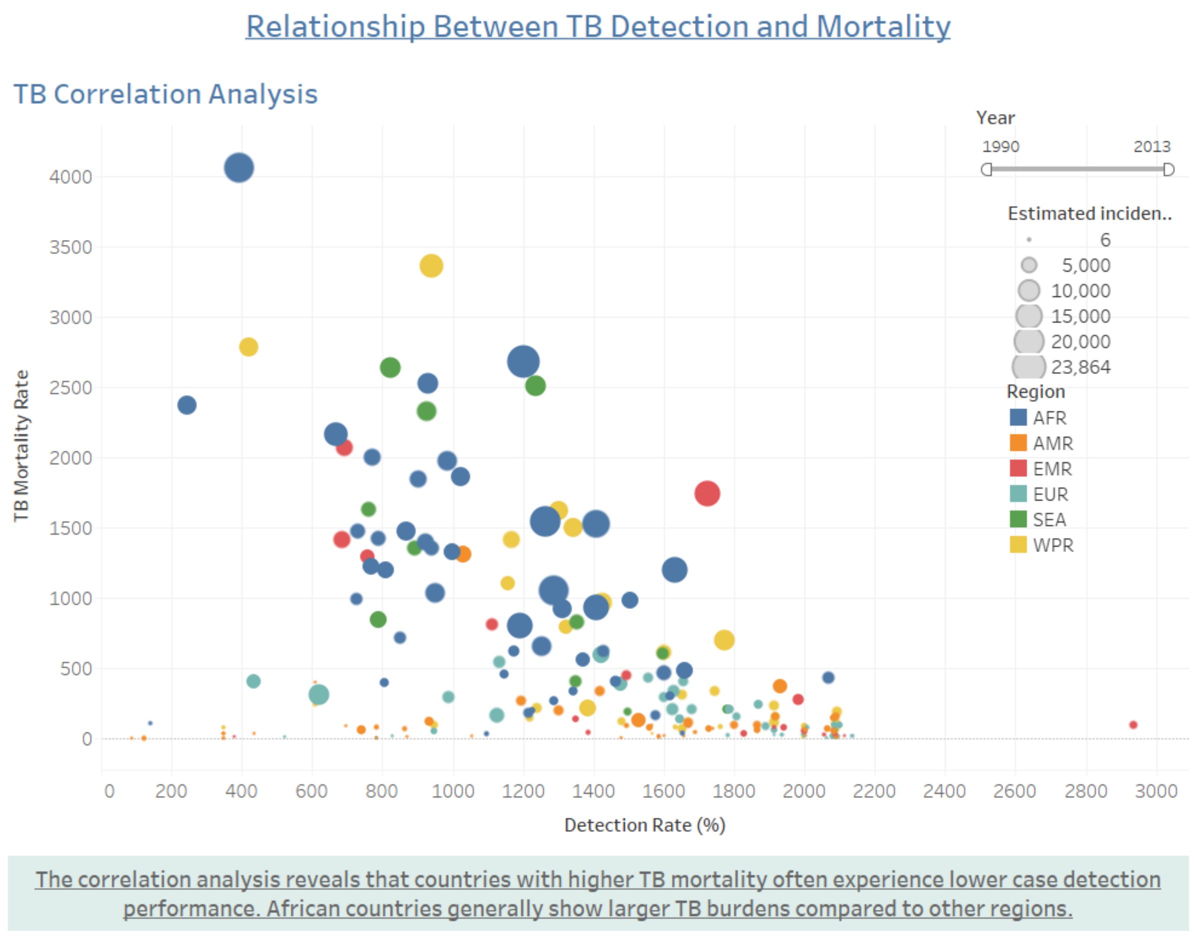

Figure 3: Relationship Between TB Detection and Mortality Dashboard

5 Storytelling Structure

The dashboards were designed to follow a structured storytelling approach:

1. The first dashboard introduces global TB trends and summarizes the overall burden through KPIs and regional comparisons.
2. The second dashboard focuses on geographic analysis, allowing users to identify countries and regions with the highest TB incidence rates.
3. The third dashboard provides deeper analytical insights by examining the relationship between TB mortality and detection performance.

Together, the dashboards create a connected narrative explaining global TB trends, regional disparities, and healthcare performance indicators.

6 KPIs, Filters, and Interactive Elements

6.1 KPIs Used

- Total Estimated Cases
- Average Detection Rate
- Highest Burden Year

6.2 Filters and Slicers

- Year Filter
- Region Filter

6.3 Interactive Features

- Interactive Tooltips
- Dynamic Filtering
- Geographic Mapping
- Bubble Size Encoding
- Region-Based Color Encoding

7 Conclusion

The analysis demonstrates that TB burden remains unevenly distributed across the world, with African countries experiencing significantly higher incidence and mortality levels compared to other regions.

The dashboards also suggest a relationship between healthcare performance and TB outcomes, where lower detection rates are often associated with higher mortality.

Through the use of interactive dashboards, geographic analysis, and correlation visualization, this project successfully highlights important global TB trends while supporting effective storytelling and data exploration.